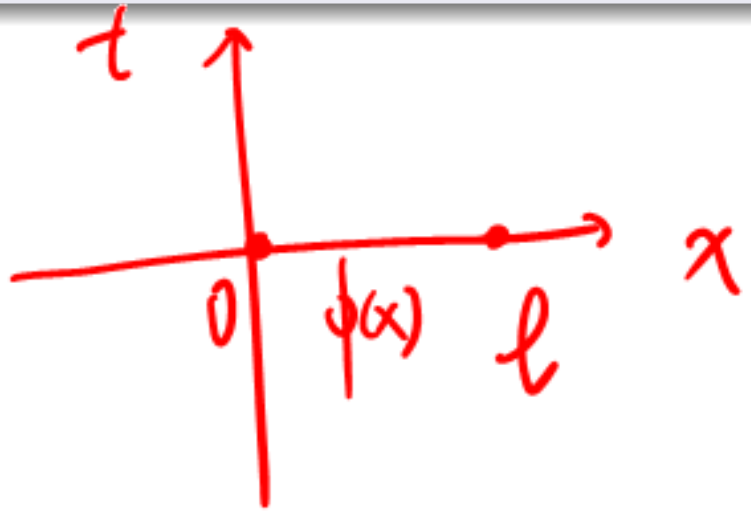
# §4.1 Separation of variables – Method

## Heat equation with homogeneous Dirichlet boundary conditions

Consider the boundary value problem

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \;\; 0 < t < \infty, \\ &u(0,t) = u(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{18}$$

where $k, l$ are positive constants and $\phi$ is a given function.



## Heat equation with homogeneous Dirichlet boundary conditions

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$$\begin{aligned} u_t &= k u_{xx}, \quad 0 < x < l, \ \ 0 < t < \infty, \\ u(0,t) &= u(l,t) = 0, \quad t > 0, \\ u(x,0) &= \phi(x), \quad 0 < x < l, \end{aligned} \tag{18}$$

$0 = X(0)T(t) = X(l)T(t)$

$\Rightarrow 0 = X(0) = X(l)$

where $k, l$ are positive constants and $\phi$ is a given function.

$X(x)T'(t) = k X''(x) T(t)$

**Separation of Variables**

(i) Look for solution of the form: $u(x,t) = X(x)T(t)$. In this case, we get

$$-\frac{X''}{X} = -\frac{T'}{kT} := \lambda.$$

$\frac{X(x)T'(t)}{-kXT} = \frac{kX''T}{-kXT}$

Once again, $\lambda$ must be a constant (why?).

$X'' + \lambda X = 0 \qquad 0 = X(0) = X(l)$

# §4.1 Separation of Variables–Method

(ii) We have the same eigenproblem as

$$X'' = -\lambda X, \quad X(0) = X(l) = 0, \tag{19}$$

which admits the eigen-pairs

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\frac{n\pi x}{l}, \quad n = 1, 2, \cdots. \tag{20}$$

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which admits the eigen-pairs

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\frac{n\pi x}{l}, \quad n = 1, 2, \cdots. \tag{20}$$

(iii) Solve $T' + \lambda k T = 0$, and obtain

$$T_n(t) = A_n \exp\left(-\left(\frac{n\pi}{l}\right)^2 kt\right).$$

$$\frac{T'}{T} = -\lambda k$$

$$d \ln|T| = -\lambda k$$

$$\ln|T| = -\lambda k t + c_1$$

$$T(t) = \pm e^{c_1} e^{-\lambda k t}$$

# §4.1 Separation of Variables–Method

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$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\frac{n\pi x}{l}, \quad n = 1, 2, \cdots. \tag{20}$$

(iii) Solve $T' + \lambda k T = 0$, and obtain

$$T_n(t) = A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big).$$

(iv) The solution is

$$u(x,t) = \sum_{n=1}^{\infty} \underbrace{A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big)}_{T_n(t)} \underbrace{\sin\frac{n\pi x}{l}}_{X_n(t)}.$$

In view of $u(x,0) = \phi(x)$, with $A_n$ to be determined we deduce that

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{l}.$$

## Theorem 2

The heat equation (18) has the following series solution:

$$u(x,t) = \sum_{n=1}^{\infty} A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big) \sin\frac{n\pi x}{l}, \tag{21}$$

provided that $\phi$ has the expansion

$$\phi(x) = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{l}. \tag{22}$$

In Chapter 5, we shall show how to find $A_n$ from $\phi$.

# §4.1 Separation of Variables–Example

## Example B (not in the textbook)

Solve the following Dirichlet problem for the heat equation by separation of variables.

$$
\begin{aligned}
&u_t = u_{xx}, \quad x \in (0, \pi/2),\\
&u(0, t) = u(\pi/2, t) = 0,\\
&u(x, 0) = 2\sin 6x.
\end{aligned}
$$

$k = 1 \qquad L = \frac{\pi}{2}$

$$\lambda_n = \left(\frac{n\pi}{\frac{\pi}{2}}\right)^2 = 4n^2$$

## Example B (not in the textbook)

Solve the following Dirichlet problem for the heat equation by separation of variables.

$$u(x,t) = \sum_{n=1}^{\infty} A_n e^{-4n^2 t} \sin 2n x$$

$$\begin{aligned} &u_t = u_{xx}, \quad x \in (0, \pi/2), \\ &u(0,t) = u(\pi/2, t) = 0, \\ &u(x,0) = 2\sin 6x. \end{aligned}$$

$$u(x,0) = \sum_{n=1}^{\infty} A_n \sin 2nx$$

**Solution:** In this problem $l = \pi/2$, so $\beta_n = 2n$. We can write the initial data in the form (34),

$$2\sin 6x = \sum_{n=1}^{\infty} A_n \sin 2nx,$$

which implies that $A_3 = 2$ and $A_n = 0$ for $n \neq 3$. Then from (33) the solution will be

$$u(x,t) = 2\exp(-36t)\sin 6x.$$

# §4.1 Separation of Variables–Example

## Example C (not in the textbook)

Consider the damped wave equation

$$\begin{aligned} &u_{tt} + u_t = u_{xx}, \quad x \in (0,1),\ t > 0, \\ &u(0,t) = u(1,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad x \in (0,1). \end{aligned} \tag{23}$$

$$X(x)\,T''(t) + X(x)\,T'(t) = X''(x)\,T(t)$$

$$-\frac{T''(t)+T'(t)}{T(t)} = -\frac{X''}{X} := \lambda$$

Eigen-problem : $X'' + \lambda X = 0 \quad X(0) = X(1) = 0$

# §4.1 Separation of Variables–Example

## Example C (not in the textbook)

Consider the damped wave equation

$$\begin{aligned} &u_{tt} + u_t = u_{xx}, \quad x \in (0,1),\ t > 0, \\ &u(0,t) = u(1,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad x \in (0,1). \end{aligned} \tag{23}$$

**Solution:** Look for the solution: $u(x,t) = X(x)T(t)$. Then we have

$$XT'' + XT' = X''T.$$

when $\lambda < \frac{1}{4}$

Dividing by $XT$, gives

$$\frac{T'' + T'}{T} = \frac{X''}{X} := -\lambda.$$

This yields the eigen-problem:

$$X'' + \lambda X = 0, \quad X(0) = X(1) = 0,$$

and the ODE for $T$ :

$$T'' + T' + \lambda T = 0.$$

$\lambda > 1/4$ $\quad \frac{-1 \pm \sqrt{4\lambda - 1}\, i}{2}$ $\quad \frac{-1 \pm \sqrt{1-4\lambda}}{2}$

# §4.1 Separation of Variables–Example (Cont'd)

The eigenvalues and eigenfunctions are

$$\lambda_n = n^2\pi^2, \quad X_n(x) = \sin(n\pi x), \quad n = 1, 2, \cdots .$$

We turn to the ODE for $T$ :

$$T_n(t) = e^{-t/2}\big(A_n \cos\omega_n t + B_n \sin\omega_n t\big), \quad \omega_n = \sqrt{n^2\pi^2 - 1/4}.$$

# §4.1 Separation of Variables–Example (Cont'd)

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$$T_n(t) = e^{-t/2}\big(A_n \cos\omega_n t + B_n \sin\omega_n t\big), \quad \omega_n = \sqrt{n^2\pi^2 - 1/4}.$$

The solution formula is

$$u(x,t) = \sum_{n=1}^{\infty} \underbrace{e^{-t/2}\big(A_n \cos\omega_n t + B_n \sin\omega_n t\big)}_{T_n(t)} \underbrace{\sin(n\pi x)}_{X_n(x)},$$

where $A_n$ and $B_n$ are determined by the initial conditions.

# §4.1 A short summary

- For the Dirichlet problems for the wave and heat equations on a finite interval, we solved them with the method of separation of variables. That is, we looked for the solution in the form of an infinite linear combination of separated solutions. This lead to infinite series, which solve the appropriate initial-boundary value problems, as long as the initial data can be expanded in corresponding series, which in the case of Dirichlet conditions are the Fourier sine series.

# §4.1 A short summary

- For the Dirichlet problems for the wave and heat equations on a finite interval, we solved them with the method of separation of variables. That is, we looked for the solution in the form of an infinite linear combination of separated solutions. This lead to infinite series, which solve the appropriate initial-boundary value problems, as long as the initial data can be expanded in corresponding series, which in the case of Dirichlet conditions are the Fourier sine series.
- It is essential to solve the eigenproblem for the operator $-\frac{d^2}{dx^2}$ in the space of functions satisfying the boundary conditions:

$$-X''(x) = \lambda X(x), \quad X(0) = X(l) = 0, \tag{24}$$

$AX = \lambda X$

whose eigenvalues and eigenfunctions are:

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\frac{n\pi x}{l}, \quad n = 1, 2, \cdots.$$

$A$ : matrix

# §4.1 A short summary

- For the Dirichlet problems for the wave and heat equations on a finite interval, we solved them with the method of separation of variables. That is, we looked for the solution in the form of an infinite linear combination of separated solutions. This lead to infinite series, which solve the appropriate initial-boundary value problems, as long as the initial data can be expanded in corresponding series, which in the case of Dirichlet conditions are the Fourier sine series.
- It is essential to solve the eigenproblem for the operator $-\frac{d^2}{dx^2}$ in the space of functions satisfying the boundary conditions:

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  whose eigenvalues and eigenfunctions are:

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\frac{n\pi x}{l}, \quad n = 1, 2, \cdots .$$

- However, unlike the operators in linear algebra, which have finitely many eigenvalues, in our case we have an infinite number of eigenvalues. This is due to the fact that the space of functions is infinite dimensional.

# §4.2 Separation of variables: The Neumann condition

## Wave equation with homogeneous Neumann boundary conditions

Solve the wave equation

$$
\begin{aligned}
&u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \quad t > 0, \\
&u_x(0,t) = u_x(l,t) = 0, \quad t > 0. \\
&u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l.
\end{aligned}
\tag{25}
$$

# §4.2 Separation of variables: The Neumann condition

## Wave equation with homogeneous Neumann boundary conditions

Solve the wave equation

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \quad t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0. \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l. \end{aligned} \tag{25}$$

Substituting the separated solution $u(x,t) = X(x)T(t)$ into the wave Neumann problem gives the same equations for $X$ and $T$ as in the Dirichlet case,

$$X'' = -\lambda X, \quad -T'' = c^2 \lambda T. \tag{26}$$

# §4.2 Separation of variables: The Neumann condition

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Solve the wave equation

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \quad t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0. \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l. \end{aligned} \tag{25}$$

Substituting the separated solution $u(x,t) = X(x)T(t)$ into the wave Neumann problem gives the same equations for $X$ and $T$ as in the Dirichlet case,

$$X'' = -\lambda X, \quad -T'' = c^2 \lambda T. \tag{26}$$

$$\begin{cases} X'' = -\lambda X \\ X'(0) = X'(l) = 0 \end{cases}$$

However, the boundary conditions now imply

$$X'(0)T(t) = X'(l)T(t) = 0, \quad \forall t \to X'(0) = X'(l) = 0.$$

Once again, $\lambda \geq 0$ gives meaningful solutions (why?).

# §4.2 Neumann Boundary Conditions (Cont'd)

Case 2: $\lambda = 0$ $\quad X(x) = c_1 x + c_2 \quad X'(x) = c_1$

Assume that $\lambda = \beta^2 \geq 0$. In this case, the general solution is

$$X(x) = C\cos\beta x + D\sin\beta x$$

$X'(0) = c_1$

$X(x) = c_2 \cdot 1$

and

$X'(\ell) = c_1$

$$X'(x) = -C\beta\sin\beta x + D\beta\cos\beta x.$$

$c_1 = 0$

Therefore, $X'(0) = 0$ implies $D = 0$, and $X'(l) = 0$ implies $C\sin\beta l = 0$.

$X'' + \lambda X = 0 \qquad X'(0) = X'(\ell) = 0$

Case 1: $\lambda < 0 \qquad X(x) = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x}$

$X'(x) = c_1\sqrt{-\lambda}\, e^{\sqrt{-\lambda}x} - c_2\sqrt{-\lambda}\, e^{-\sqrt{-\lambda}x}$

$X'(0) = 0 \Rightarrow c_1 = c_2 \qquad X'(\ell) = 0 \Rightarrow c_1\sqrt{-\lambda}\left(e^{\sqrt{-\lambda}\ell} - e^{-\sqrt{-\lambda}\ell}\right) = 0$

Case 3: $\lambda > 0$

Assume that $\lambda = \beta^2 \geq 0$. In this case, the general solution is

$$X(x) = C\cos\beta x + D\sin\beta x$$

$X'(0) = D\beta$ $\quad D=0 \rightarrow X'(l) = -C\beta\sin\beta l$

and

$$X'(x) = -C\beta\sin\beta x + D\beta\cos\beta x.$$

Therefore, $X'(0) = 0$ implies $D = 0$, and $X'(l) = 0$ implies $C\sin\beta l = 0$.

Since we do not want $C = 0$, we have to take $\beta l = n\pi$ for any integer $n$. This leads to the eigen-pairs of (26):

$\sin\beta l = 0$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \cos\frac{n\pi x}{l}, \quad n = 0, 1, 2, \cdots, \tag{27}$$

as opposed to the Dirichlet boundary condition case. Also notice that $\lambda = 0$ is also an eigenvalue of (26) with the corresponding eigenfunction 1.

$X'' + aX = 0 \quad X'(0) = X'(l) = 0$

# §4.2 Neumann Boundary Conditions (Cont'd)

For $T(t)$,

$$T'' = -c^2 \lambda T, \quad \lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad n = 0, 1, 2, \cdots .$$

# §4.2 Neumann Boundary Conditions (Cont'd)

For $T(t)$,

$$T'' = -c^2 \lambda T, \quad \lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad n = 0, 1, 2, \cdots .$$

When $n = 0$, $T_0'' = 0$, so

$$T_0(t) = \frac{A_0}{2} + \frac{B_0}{2} t,$$

where $A_0$ and $B_0$ are constants to be determined by the initial condition. Here, the factor $1/2$ is conventional for the formulation of Fourier series (see Chapter 5).

# §4.2 Neumann Boundary Conditions (Cont'd)

For $T(t)$,

$$T'' = -c^2 \lambda T, \quad \lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad n = 0, 1, 2, \cdots .$$

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where $A_0$ and $B_0$ are constants to be determined by the initial condition. Here, the factor $1/2$ is conventional for the formulation of Fourier series (see Chapter 5).

When $n \geq 1$, the expressions of $T_n(t)$ are given by

$$T_n(t) = A_n \cos \frac{n\pi ct}{l} + B_n \sin \frac{n\pi ct}{l}. \tag{28}$$

## Theorem 3

The wave equation

$$\begin{aligned} &u_{tt} = c^2 u_{xx}, \quad 0 < x < l, \ t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad 0 < x < l, \end{aligned} \tag{29}$$

has the following series solution: $X_0(x)$

$$u(x,t) = \underbrace{\left(\frac{A_0}{2} + \frac{B_0}{2}t\right)}_{T_0(t)} + \sum_{n=1}^{\infty} \underbrace{\left(A_n \cos\frac{n\pi ct}{l} + B_n \sin\frac{n\pi ct}{l}\right)}_{T_n(t)} \underbrace{\cos\frac{n\pi x}{l}}_{X_n(x)}, \tag{30}$$

as long as the initial data can be expanded into cosine Fourier series

$$\phi(x) = \frac{1}{2}A_0 + \sum_n A_n \cos\frac{n\pi x}{l}, \quad \psi(x) = \frac{1}{2}B_0 + \sum_n \frac{n\pi c}{l} B_n \cos\frac{n\pi x}{l}. \tag{31}$$

# §4.1 Separation of Variables–Example

## Example D (not in the textbook)

Solve the following Neumann problem for the wave equation by separation of variables.

$$
\begin{aligned}
&u_{tt} - 4u_{xx} = 0, \quad x \in (0, \pi), \\
&u_x(0, t) = u_x(\pi, t) = 0, \\
&u(x, 0) = 3\cos x, \quad u_t(x, 0) = 1 - \cos 4x.
\end{aligned}
$$

$L = \pi$

$c = 2$

## Example D (not in the textbook)

Solve the following Neumann problem for the wave equation by separation of variables.

$$
\begin{aligned}
&u_{tt} - 4u_{xx} = 0, \quad x \in (0, \pi), \\
&u_x(0, t) = u_x(\pi, t) = 0, \\
&u(x, 0) = 3\cos x, \quad u_t(x, 0) = 1 - \cos 4x.
\end{aligned}
$$

**Solution:** In this problem $l = \pi$, so $\beta_n = n$. Notice that $c = 2$, and we can write the initial data in the form (31) as follows,

$$
3\cos x = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos nx, \quad 1 - \cos 4x = \frac{1}{2}B_0 + \sum_{n=1}^{\infty} 2nB_n \cos nx.
$$

$$
\begin{cases} A_1 = 3 \\ A_n = 0 \quad n \neq 1 \end{cases}
$$

$B_0 = 2$

$B_n = 0 \quad n \neq 4$

$8B_4 = -1 \Rightarrow B_4 = -1/8$

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Solve the following Neumann problem for the wave equation by separation of variables.

$$\begin{aligned} &u_{tt} - 4u_{xx} = 0, \quad x \in (0, \pi), \\ &u_x(0, t) = u_x(\pi, t) = 0, \\ &u(x, 0) = 3\cos x, \quad u_t(x, 0) = 1 - \cos 4x. \end{aligned}$$

**Solution:** In this problem $l = \pi$, so $\beta_n = n$. Notice that $c = 2$, and we can write the initial data in the form (31) as follows,

$$3\cos x = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos nx, \quad 1 - \cos 4x = \frac{1}{2}B_0 + \sum_{n=1}^{\infty} 2nB_n \cos nx.$$

These identities imply that $A_1 = 3$, and $A_n = 0$ for $n \neq 1$, and $B_0 = 2$, $B_4 = -1/8$, and $B_n = 0$ for $n \neq 0, 4$.

# §4.1 Separation of Variables–Example

## Example D (not in the textbook)

Solve the following Neumann problem for the wave equation by separation of variables.

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**Solution:** In this problem $l = \pi$, so $\beta_n = n$. Notice that $c = 2$, and we can write the initial data in the form (31) as follows,

$$3\cos x = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos nx, \quad 1 - \cos 4x = \frac{1}{2}B_0 + \sum_{n=1}^{\infty} 2nB_n \cos nx.$$

These identities imply that $A_1 = 3$, and $A_n = 0$ for $n \neq 1$, and $B_0 = 2$, $B_4 = -1/8$, and $B_n = 0$ for $n \neq 0, 4$. Therefore, the solution is

$$u(x, t) = t + 3\cos 2t \cos x - \frac{1}{8}\sin 8t \cos 4x.$$

## Theorem 4

The heat equation

$$\begin{aligned} &u_t = ku_{xx}, \quad 0 < x < l, \ t > 0, \\ &u_x(0,t) = u_x(l,t) = 0, \quad t > 0, \\ &u(x,0) = \phi(x), \quad 0 < x < l, \end{aligned} \tag{32}$$

has the series solution:

$$u(x,t) = \underbrace{\frac{1}{2}A_0}_{X_0 T_0} + \sum_{n=1}^{\infty} \underbrace{A_n \exp\Big(-\Big(\frac{n\pi}{l}\Big)^2 kt\Big)}_{T_n(t)} \underbrace{\cos\frac{n\pi x}{l}}_{X_n(x)}, \tag{33}$$

provided that $\phi$ has the cosine Fourier expansion

$$\phi(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l}. \tag{34}$$

In Chapter 5, we shall show how to find $A_n$ from $\phi$.

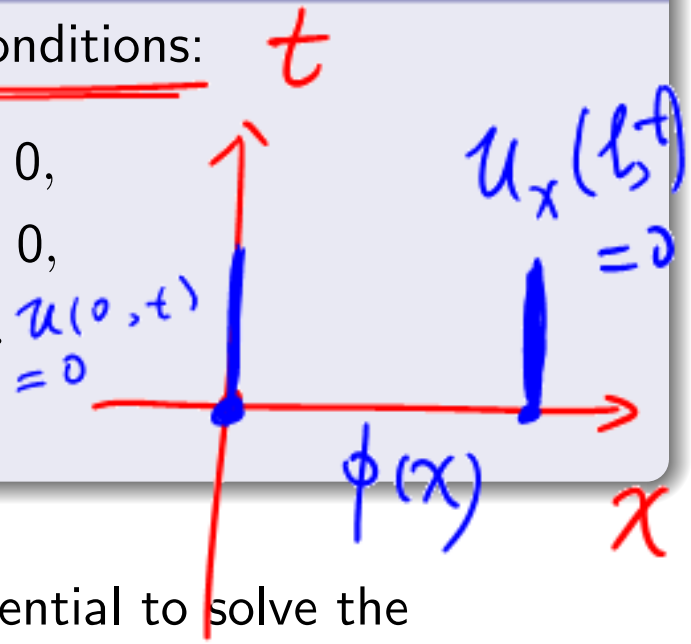
# §4.2 Mixed Boundary Conditions– Example

Sometimes one needs to consider problems with mixed Dirichlet-Neumann boundary conditions, i.e. Dirichlet conditions at one end of the finite interval, and Neumann conditions at the other.

## Example E

Consider the heat equation with "mixed" boundary conditions:

$$\begin{aligned} u_t &= ku_{xx}, \quad 0 < x < l, \ t > 0, \\ u(0,t) &= u_x(l,t) = 0, \quad t > 0, \\ u(x,0) &= \phi(x), \quad 0 < x < l. \end{aligned}$$




Derive the solution formula.

**Solution:** Assume that $u(x,t) = X(x)T(t)$. It is essential to solve the eigenproblem:

$$X'' + \lambda X = 0, \quad X(0) = X'(l) = 0.$$

Case 1: $\lambda < 0$ $\quad X(x) = c_1 e^{\sqrt{-\lambda}\, x} + c_2 e^{-\sqrt{-\lambda}\, x} \Rightarrow c_1 = c_2 = 0$

# §4.2 Mixed Boundary Conditions– Example (Cont'd)

- For $\lambda > 0$, we find that $X(0) = C$

$$X(x) = C\cos\sqrt{\lambda}x + D\sin\sqrt{\lambda}x.$$

$\rightarrow$ So, $X'(x) = D\sqrt{\lambda}\cos\sqrt{\lambda}x$

Therefore, $X(0)$ implies $C = 0$, and $X'(l) = D\sqrt{\lambda}\cos\sqrt{\lambda}l = 0$. Since we do not want $D$ to be zero, we have to take

$\sqrt{\lambda}l = n\pi + \frac{\pi}{2}$

$$\lambda_n = \left(n + \frac{1}{2}\right)^2\frac{\pi^2}{l^2}, \quad X_n(x) = \sin\left(\left(n + \frac{1}{2}\right)\frac{\pi x}{l}\right),$$

for $n = 0, 1, \cdots$.

Case 2: $\lambda = 0$ $\quad X''(x) = 0 \quad X(0) = X'(l) = 0$

$$X(x) = C_1 x + C_2 \xRightarrow{X(0)=0} C_2 = 0$$

$$X'(x) = C_1 \xRightarrow{X'(l)=0} C_1 = 0$$

# §4.2 Mixed Boundary Conditions– Example (Cont'd)

- For $\lambda > 0$, we find that

$$X(x) = C\cos\sqrt{\lambda}x + D\sin\sqrt{\lambda}x.$$

Therefore, $X(0)$ implies $C = 0$, and $X'(l) = D\sqrt{\lambda}\cos\sqrt{\lambda}l = 0$. Since we do not want $D$ to be zero, we have to take

$$\lambda_n = \left(n + \frac{1}{2}\right)^2 \frac{\pi^2}{l^2}, \quad X_n(x) = \sin\left(\left(n + \frac{1}{2}\right)\frac{\pi x}{l}\right),$$

for $n = 0, 1, \cdots$ .

- For $\lambda = 0$, we see that $X'' = 0$ and $X(x) = ax + b$. Hence, $X(0) = b = 0$ and $X'(l) = a = 0$. This gives trivial solution $u = 0$, which should be rejected.

# §4.2 Mixed Boundary Conditions– Example (Cont'd)

- For $\lambda > 0$, we find that

$$X(x) = C\cos\sqrt{\lambda}x + D\sin\sqrt{\lambda}x.$$

Therefore, $X(0)$ implies $C = 0$, and $X'(l) = D\sqrt{\lambda}\cos\sqrt{\lambda}l = 0$. Since we do not want $D$ to be zero, we have to take

$$\lambda_n = \left(n + \frac{1}{2}\right)^2 \frac{\pi^2}{l^2}, \quad X_n(x) = \sin\left(\left(n + \frac{1}{2}\right)\frac{\pi x}{l}\right),$$

for $n = 0, 1, \cdots$ .

- For $\lambda = 0$, we see that $X'' = 0$ and $X(x) = ax + b$. Hence, $X(0) = b = 0$ and $X'(l) = a = 0$. This gives trivial solution $u = 0$, which should be rejected.

- For $\lambda < 0$, we find

$$X(x) = Ce^{-\sqrt{-\lambda}x} + De^{\sqrt{-\lambda}x}.$$

By requiring $X(0) = X'(l) = 0$, we can show that $C = D = 0$. Therefore, $\lambda < 0$ is not possible.

# §4.3 Mixed Boundary Conditions– Example (Cont'd)

$$T'(t) + \lambda k T(t) = 0$$

The equation for $T(t)$ is still the same. Therefore, we obtain the solution formula:

$$u(x,t) = \sum_{n=0}^{\infty} \underbrace{A_n \exp\Big(-\Big(n+\frac{1}{2}\Big)^2 \frac{\pi^2}{l^2} kt\Big)}_{T_n(t)} \underbrace{\sin\Big(\Big(n+\frac{1}{2}\Big)\frac{\pi x}{l}\Big)}_{X_n(x)}, \tag{35}$$

where again $A_n$'s are determined by $\phi$.

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where again $A_n$'s are determined by $\phi$.

If the boundary condition is alternatively as $u_x(0,t) = u(l,t) = 0, \quad t > 0$, then the solution formula is

$$u(x,t) = \sum_{n=0}^{\infty} A_n \exp\Big(-\Big(n+\frac{1}{2}\Big)^2 \frac{\pi^2}{l^2} kt\Big) \cos\Big(\Big(n+\frac{1}{2}\Big)\frac{\pi x}{l}\Big), \tag{36}$$

# A short summary

- Similar to the case of the Dirichlet problems for heat and wave equations, the method of separation of variables applied to the Neumann problems on a finite interval leads to an eigenvalue problem for the $X(x)$ factor of the separated solution.

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- In this case, however, we discovered a new eigenvalue $\lambda = 0$ in addition to the eigenvalues found for the Dirichlet problems.
- The general solutions of the Neumann problems for wave and heat equations can be written in series forms, as (infinite) linear combinations of all separated solutions, as long as the initial data can be expanded in cosine Fourier series.

# A short summary

We have seen that separation of variables essentially depends on solving the eigen-problem $X'' + \lambda X = 0$ with different types of boundary conditions. Previously, we just discussed the Dirichlet, Neumann and special "mixed" boundary conditions.

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We have seen that separation of variables essentially depends on solving the eigen-problem $X'' + \lambda X = 0$ with different types of boundary conditions. Previously, we just discussed the Dirichlet, Neumann and special "mixed" boundary conditions.

Indeed, the method could work for $X(x) = C_1 \cos\sqrt{\lambda}\, x + C_2 \sin\sqrt{\lambda}\, x$

- more general "mixed" boundary conditions (see Chapter 5);
- inhomogeneous boundary conditions and source terms (see Chapter 5);

$\sin\sqrt{\lambda}\, a = \sin\sqrt{\lambda}\, b = 0$

$\Downarrow$

- other types of equations, e.g., the Laplace's equation (we will discuss this in Chapter 6, if time allows):

$$\Delta u = u_{xx} + u_{yy} = 0;$$

$\sqrt{\lambda}\, a = k_1 \pi \qquad b > a$

$\sqrt{\lambda}\, b = k_2 \pi \qquad$ so $k_2 > k_1$

$\sqrt{\lambda} = (k_2 - k_1)\pi / b - a$

$$\begin{cases} C_1 \cos\sqrt{\lambda}\, a + C_2 \sin\sqrt{\lambda}\, a = 0 \\ C_1 \cos\sqrt{\lambda}\, b + C_2 \sin\sqrt{\lambda}\, b = 0 \end{cases}$$

- other types of separable domains, other than $(0, l) \times (0, \infty)$, e.g., ball and disk, but they are usually much more complicated.

Assume $C_1 = 0$, $C_2 \neq 0$ then $\sin\sqrt{\lambda}\, a = \sin\sqrt{\lambda}\, b = 0$

# Boundary value problems

Let $\xi = \dfrac{x-a}{b-a}$ $\quad x=a,\ \xi=0;\ x=b,\ \xi=1$

Exercise: Find the solution for the wave equation

$$u_{tt} = c^2 u_{xx}, \quad a < x < b, \quad t > 0,$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\frac{d\xi}{dx} = \frac{1}{b-a} u_\xi$$

subjected to the homogeneous boundary conditions

$$u(a,t) = u(b,t) = 0, \quad t > 0,$$

and the initial condition

$$u_{xx} = \frac{1}{(b-a)^2} u_{\xi\xi}$$

$$u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \quad a < x < b,$$

where $c > 0$ and $a, b$ are given constants, and $\phi, \psi$ are given functions.

$$u_{tt} = \frac{c^2}{(b-a)^2} u_{\xi\xi}, \quad 0 < \xi < 1, \ t > 0$$

$$u(\xi=0, t) = u(\xi=1, t) = 0$$